How Valuable Is Financial Flexibility when Revenue Stops? Evidence from the COVID-19 Crisis

Replication Results

Extended sample

Fahlenbrach, R., K. Rageth, and R. M. Stulz (2021), *Review of Financial Studies* 34(11)

Anthony Lamini

Advanced Financial Theorem

October 17, 2026

Appendix A
Sample selection procedure

Extended sample. Compustat North America firms with fiscal year 2019 data, after each screen.

Number of firms with fiscal year 2019 data	11,612
Duplicate data	−1
SIC codes 6000-6999 (financial firms)	−4,872
SIC codes 4900-4949 (utility firms)	−264
SIC codes 8000s and 9000s (not-for-profit and governmental firms)	−449
Non-U.S. headquarters	−2,195
Missing cash and short-term investments	−166
Missing stock price data in Security Daily	−501
An average share price $<$ \$1	−410
Security type not equal to “common, ordinary”	−26
Number of firms after all screens	2,729

Table C1
Summary statistics

Summary statistics for the raw data and the derived measures of COVID-19 exposure of firms and industries. TQ (BQ) indicates an indicator for the top (bottom) quartile.

	N	P25	Median	Mean	P75	SD
Affected share	2,701	9.000	10.000	15.582	14.000	15.335
Teamwork share	2,701	17.000	25.000	24.906	25.000	10.729
Customer share	2,701	7.000	8.000	14.922	17.000	15.894
Presence share	2,701	3.000	9.000	10.431	11.000	12.427
Q2 2020 year-on-year change in sales	2,439	−0.349	−0.110	0.439	0.090	9.910
Share of jobs that can be performed from home	2,719	0.210	0.278	0.380	0.506	0.236
Labor intensity as number of employees / sales	2,474	0.002	0.003	0.018	0.005	0.245
Affected share (TQ)	2,701	0.000	0.000	0.247	0.000	0.431
Teamwork share (TQ)	2,701	0.000	0.000	0.226	0.000	0.418
Customer share (TQ)	2,701	0.000	0.000	0.230	0.000	0.421
Presence share (TQ)	2,701	0.000	0.000	0.190	0.000	0.392
Q2 2020 year-on-year change in sales (BQ)	2,439	0.000	0.000	0.250	0.500	0.433
Share of jobs that can be performed from home (BQ)	2,719	0.000	0.000	0.252	1.000	0.434
Labor intensity based on number of employees (TQ)	2,474	0.000	0.000	0.250	0.750	0.433
Manually classified as highly affected	2,701	0.000	0.000	0.245	0.000	0.430

Table C2

Correlation matrix

Correlations among the measures of COVID-19 exposure of firms and industries.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) Affected share (TQ)	1.00							
(2) Teamwork share (TQ)	0.13	1.00						
(3) Customer share (TQ)	0.55	0.24	1.00					
(4) Presence share (TQ)	0.39	0.11	−0.07	1.00				
(5) Q2 2020 year-on-year change in sales (BQ)	0.17	0.04	0.01	0.11	1.00			
(6) Share of jobs that can be performed from home (BQ)	0.22	−0.18	0.12	0.24	0.09	1.00		
(7) Labor intensity based on number of employees (TQ)	0.04	−0.09	0.13	−0.18	0.06	0.01	1.00	
(8) Manually classified as highly affected	0.97	0.16	0.52	0.42	0.20	0.22	0.04	1

Table 1
Summary statistics

Summary statistics for the dependent and independent variables. The sample consists of all nonfinancial and nonutility U.S. firms with fiscal year 2019 data in Compustat that pass the screens in Appendix A. GFC, global financial crisis.

	N	Min	P1	P5	P25	Median	Mean	P75	P95	P99	Max	SD
Dependent variables: COVID-19-related												
Cum. daily exc. returns (Feb 3 - Mar 23, 2020)	2,729	−0.974	−0.824	−0.711	−0.524	−0.381	−0.366	−0.254	−0.001	0.457	3.534	0.273
Daily exc. returns (March 24, 2020)	2,717	−0.481	−0.125	−0.039	0.036	0.084	0.091	0.135	0.257	0.395	1.165	0.098
ROA (cumulative Q1 + Q2 2020)	2,596	−1.300	−1.296	−0.425	−0.067	0.021	−0.048	0.056	0.116	0.214	0.214	0.214
Dependent variables: GFC-related												
Cum. daily exc. returns (Sep 15 - Dec 31, 2008)	3,179	−0.998	−0.939	−0.824	−0.589	−0.407	−0.393	−0.231	0.031	0.385	11.662	0.368
Cum. daily exc. returns (Sep 15, 2008 - Mar 9, 2009)	3,179	−0.999	−0.974	−0.904	−0.728	−0.551	−0.510	−0.370	−0.001	0.475	9.328	0.370
Independent and other variables												
Assets ($m)	2,729	0.002	2.591	12.477	116.250	682.837	6,168.997	2,901.765	24,694.680	100,639.680	551,669.000	25,055.867
Cash / assets	2,729	0.000	0.001	0.006	0.041	0.130	0.257	0.402	0.889	0.972	0.972	0.285
St debt / assets	2,729	0.000	0.000	0.000	0.005	0.015	0.047	0.042	0.201	0.676	0.677	0.099
nSt debt / assets	2,729	−0.968	−0.968	−0.875	−0.365	−0.093	−0.209	−0.007	0.112	0.631	0.631	0.315
Lt debt / assets	2,722	0.000	0.000	0.000	0.047	0.213	0.253	0.385	0.686	1.179	1.182	0.238
Book debt / assets	2,722	0.000	0.000	0.000	0.085	0.262	0.307	0.445	0.768	1.586	1.588	0.281
Payout / assets	2,729	0.000	0.000	0.000	0.000	0.007	0.033	0.036	0.154	0.383	0.383	0.062
Ind: 3-year cum. payout / assets top quartile	2,649	0.000	0.000	0.000	0.000	0.000	0.250	0.000	1.000	1.000	1.000	0.433
Ind: COVID-19 conglomerate	2,729	0.000	0.000	0.000	0.000	0.000	0.083	0.000	1.000	1.000	1.000	0.276
Ind: IG debt rating	2,729	0.000	0.000	0.000	0.000	0.000	0.109	0.000	1.000	1.000	1.000	0.311
Ind: HY debt rating	2,729	0.000	0.000	0.000	0.000	0.000	0.150	0.000	1.000	1.000	1.000	0.357
RD / assets	2,729	0.000	0.000	0.000	0.000	0.013	0.111	0.112	0.507	1.071	7.161	0.290
SGA / sales	2,282	0.021	0.022	0.048	0.128	0.256	0.670	0.490	1.665	16.366	16.369	1.952
Capex / lagged assets	2,698	0.000	0.000	0.001	0.011	0.026	0.043	0.053	0.152	0.306	0.306	0.053
COGS / sales	2,538	0.000	0.052	0.169	0.429	0.635	0.609	0.798	1.000	1.000	1.000	0.250
KZ index	2,600	−41,770.963	−1,228.498	−139.139	−12.030	−2.240	−53.353	1.016	6.300	110.754	42,794.399	1,302.618
FC (KZ index)	2,600	0.000	0.000	0.000	0.000	0.000	0.250	0.250	1.000	1.000	1.000	0.433
WW index	2,496	−140.912	−0.553	−0.488	−0.375	−0.263	−0.167	−0.106	0.172	7.360	7.629	2.999
FC (WW index)	2,496	0.000	0.000	0.000	0.000	0.000	0.250	0.250	1.000	1.000	1.000	0.433
SA index	2,729	−4.637	−4.637	−4.637	−4.117	−3.443	−3.403	−2.910	−1.807	−0.599	6.550	0.960
FC (SA index)	2,729	0.000	0.000	0.000	0.000	0.000	0.250	0.000	1.000	1.000	1.000	0.433
HR cash ex ante	2,701	−20,905.000	−1,326.676	−134.976	1.838	59.683	812.766	343.398	2,621.000	11,523.000	211,611.000	6,708.533
FC (HR cash ex ante)	2,701	0.000	0.000	0.000	0.000	0.000	0.222	0.000	1.000	1.000	1.000	0.416
Equity beta	2,698	−3.184	−0.438	0.054	0.680	1.054	1.037	1.427	1.972	2.397	4.227	0.609
BM at EoY 2019	2,723	−1.819	−1.806	−0.109	0.145	0.341	0.496	0.691	1.762	3.508	3.509	0.692
ln(MVE) at EoY 2019	2,728	−1.884	1.589	2.594	4.850	6.622	6.529	8.144	10.361	11.879	11.879	2.336
Momentum 2019	2,727	−0.988	−0.883	−0.685	−0.163	0.140	0.200	0.435	1.187	3.065	3.084	0.630
Profitability	2,729	−6.900	−1.110	−0.362	0.117	0.243	0.226	0.392	0.700	1.176	1.179	0.390

Table 2
Correlation matrix

Correlations among the dependent and independent variables for the sample described in Table 1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)
(1) Cum. daily exc. returns (Feb 3 - Mar 23, 2020)	1.00																								
(2) Daily exc. returns (March 24, 2020)	−0.34	1.00																							
(3) ROA (cumulative Q1 + Q2 2020)	−0.04	0.12	1.00																						
(4) Assets ($m)	0.01	0.04	0.11	1.00																					
(5) Cash / assets	0.17	−0.09	−0.49	−0.13	1.00																				
(6) St debt / assets	−0.03	−0.07	−0.26	−0.01	−0.11	1.00																			
(7) nSt debt / assets	−0.16	0.05	0.35	0.11	−0.93	0.45	1.00																		
(8) Lt debt / assets	−0.16	0.15	0.19	0.07	−0.34	0.01	0.31	1.00																	
(9) Book debt / assets	−0.15	0.09	0.03	0.05	−0.32	0.46	0.46	0.87	1.00																
(10) Payout / assets	0.00	0.06	0.17	0.10	−0.09	0.01	0.09	0.08	0.09	1.00															
(11) Ind: 3-year cum. payout / assets top quartile	0.02	0.05	0.18	0.13	−0.13	−0.02	0.11	0.07	0.05	0.65	1.00														
(12) Ind: COVID-19 conglomerate	−0.02	0.00	0.11	0.07	−0.14	0.00	0.12	0.06	0.05	0.01	0.04	1.00													
(13) Ind: IG debt rating	−0.01	0.08	0.17	0.45	−0.19	−0.02	0.16	0.09	0.06	0.22	0.30	0.09	1.00												
(14) Ind: HY debt rating	−0.17	0.12	0.16	0.01	−0.26	−0.02	0.23	0.33	0.27	−0.03	−0.04	0.05	−0.15	1.00											
(15) RD / assets	0.12	−0.10	−0.65	−0.07	0.44	0.11	−0.35	−0.16	−0.05	−0.09	−0.10	−0.10	−0.11	−0.14	1.00										
(16) SGA / sales	0.09	−0.07	−0.58	−0.06	0.35	0.12	−0.25	−0.10	−0.02	−0.05	−0.07	−0.06	−0.09	−0.11	0.42	1.00									
(17) Capex / lagged assets	−0.12	0.08	0.12	0.03	−0.23	−0.03	0.19	0.11	0.08	0.00	0.00	0.03	0.02	0.05	−0.15	−0.09	1.00								
(18) COGS / sales	−0.03	−0.02	−0.17	−0.05	0.01	0.13	0.03	−0.03	0.02	−0.16	−0.11	0.06	−0.05	0.05	0.07	−0.13	−0.05	1.00							
(19) FC (KZ index)	−0.09	0.03	−0.18	−0.06	−0.21	0.23	0.26	0.32	0.38	−0.19	−0.20	−0.02	−0.13	0.15	0.07	0.09	0.19	0.13	1.00						
(20) FC (WW index)	0.13	−0.13	−0.47	−0.13	0.47	0.12	−0.37	−0.19	−0.10	−0.12	−0.14	−0.13	−0.18	−0.20	0.36	0.33	−0.19	0.10	0.04	1.00					
(21) FC (SA index)	0.12	−0.18	−0.55	−0.14	0.48	0.17	−0.38	−0.30	−0.16	−0.12	−0.15	−0.12	−0.20	−0.23	0.37	0.36	−0.14	0.09	0.10	0.50	1.00				
(22) FC (HR cash ex ante)	−0.04	−0.09	−0.29	−0.05	−0.15	0.19	0.20	0.08	0.17	−0.06	−0.07	0.00	−0.09	−0.03	0.16	0.19	0.18	0.08	0.29	0.11	0.18	1.00			
(23) Equity beta	−0.09	0.14	0.05	0.00	0.08	−0.15	−0.13	0.09	−0.01	−0.06	−0.09	−0.04	0.01	0.13	0.00	−0.04	0.09	−0.04	0.00	−0.05	−0.20	−0.06	1.00		
(24) BM at EoY	−0.04	−0.05	0.08	−0.03	−0.15	−0.10	0.10	−0.21	−0.24	−0.13	−0.15	0.05	−0.07	0.04	−0.13	−0.05	0.08	0.13	0.05	−0.08	−0.07	0.01	−0.02	1.00	
(25) ln(MVE) at EoY	−0.04	0.23	0.39	0.40	−0.20	−0.21	0.12	0.22	0.09	0.25	0.30	0.08	0.51	0.16	−0.20	−0.20	0.10	−0.23	−0.26	−0.40	−0.55	−0.24	0.28	−0.26	1.00
(26) Momentum	−0.01	0.09	0.07	0.02	0.08	−0.10	−0.10	−0.02	−0.04	0.01	0.00	0.01	0.04	0.00	−0.04	−0.07	−0.01	−0.07	−0.15	−0.02	−0.07	−0.16	0.02	−0.23	0.27
(27) Profitability	−0.03	0.07	0.49	0.02	−0.32	−0.09	0.26	0.03	−0.02	0.19	0.17	0.05	0.06	0.03	−0.43	−0.23	0.06	−0.47	−0.13	−0.26	−0.24	−0.19	−0.03	−0.01	0.15

Table 3

Stock returns, financial flexibility measures, and stock and firm characteristics

Cross-sectional regressions of stock returns in excess of the risk-free rate on firm characteristics. Collapse period: cumulative return from February 3 to March 23, 2020. Stimulus day: return on March 24, 2020. Columns 7 and 8 use the global financial crisis, with cumulative returns from September 15, 2008, to December 31, 2008, and to March 9, 2009, and firm characteristics measured in fiscal year 2006. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $*p < .1$; $**p < .05$; $***p < .01$.

	(1) Collapse period	(2) Stimulus day	(3) Collapse period	(4) Stimulus day	(5) Collapse period	(6) Stimulus day	(7) Global financial crisis ending Dec 31, 2008	(8) Global financial crisis ending March 9, 2009
Cash / assets	0.038	0.016			0.040	0.015	0.123***	0.159***
	(1.08)	(1.20)			(1.13)	(1.20)	(2.87)	(3.87)
St debt / assets	−0.105	−0.002					−0.208**	−0.310***
	(−1.61)	(−0.09)					(−2.07)	(−3.20)
nSt debt / assets			−0.049*	−0.016				
			(−1.76)	(−1.51)				
Lt debt / assets	−0.149***	0.033***	−0.117***	0.025**			−0.114***	−0.066
	(−5.36)	(3.19)	(−4.11)	(2.39)			(−2.66)	(−1.60)
Book debt / assets					−0.130***	0.023**		
					(−5.31)	(2.55)		
Ind: IG debt rating	0.022	−0.014*	−0.002	−0.008	0.023	−0.015**	0.053*	0.051*
	(1.09)	(−1.91)	(−0.10)	(−1.02)	(1.16)	(−2.05)	(1.82)	(1.82)
Ind: HY debt rating			−0.058***	0.015**				
			(−3.56)	(2.48)				
Capex / lagged assets	−0.106	0.064	−0.121	0.068	−0.107	0.064	−0.018	−0.218***
	(−0.91)	(1.49)	(−1.04)	(1.60)	(−0.93)	(1.50)	(−0.25)	(−3.12)
SGA / sales	−0.003	0.001	−0.003	0.001	−0.003	0.001	−0.000	0.002
	(−0.77)	(0.85)	(−0.85)	(0.86)	(−0.82)	(0.88)	(−0.06)	(0.39)
COGS / sales	0.001	0.007	0.010	0.005	0.006	0.005	−0.000	−0.015
	(0.02)	(0.60)	(0.31)	(0.44)	(0.19)	(0.40)	(−0.01)	(−0.35)
RD / assets	0.028	−0.008	0.024	−0.007	0.046	−0.013	0.012	0.233***
	(0.85)	(−0.68)	(0.73)	(−0.55)	(1.40)	(−1.12)	(0.19)	(3.73)
Payout / assets	0.096	0.030	0.075	0.035	0.101	0.028	0.122	0.182*
	(1.02)	(0.88)	(0.80)	(1.02)	(1.07)	(0.82)	(1.18)	(1.82)
Equity beta	−0.059***	0.015***	−0.057***	0.014***	−0.060***	0.015***	−0.014	−0.053***
	(−5.42)	(3.64)	(−5.25)	(3.49)	(−5.55)	(3.80)	(−1.16)	(−4.53)
BM at EoY 2019	−0.000	−0.001	0.004	−0.002	−0.000	−0.001	−0.032	−0.066***
	(−0.05)	(−0.18)	(0.46)	(−0.51)	(−0.04)	(−0.19)	(−1.43)	(−3.09)
ln(MVE) at EoY 2019	0.002	0.008***	0.005	0.007***	0.001	0.009***	0.014**	0.005
	(0.50)	(6.34)	(1.48)	(5.77)	(0.34)	(6.83)	(2.54)	(0.92)
Momentum 2019	−0.021**	0.009**	−0.022**	0.009**	−0.021**	0.009**	0.012	−0.008
	(−2.02)	(2.30)	(−2.06)	(2.32)	(−1.96)	(2.28)	(0.93)	(−0.64)
Profitability	−0.040	0.002	−0.037	0.002	−0.037	0.002	−0.012	−0.024
	(−1.43)	(0.22)	(−1.35)	(0.17)	(−1.32)	(0.18)	(−0.31)	(−0.66)
Observations	2,237	2,232	2,237	2,232	2,237	2,232	2,766	2,766
Adjusted R^2	0.132	0.111	0.137	0.113	0.132	0.110	0.035	0.080

Table 4

Bottom-20 NAICS three-digit industries cumulative returns and industry characteristics: GFC versus COVID-19 crisis

The 20 NAICS three-digit industries with the lowest equal-weighted cumulative excess return (CR) from September 15, 2008, to March 9, 2009 (left) and from February 3 to March 23, 2020 (right). * marks industries in the bottom 20 in both crises.

Global financial crisis					COVID-19 crisis				
NAICS	Affected share	Lt debt / assets	PPE / assets	CR	NAICS	Affected share	Lt debt / assets	PPE / assets	CR
* 314	10.000	0.245	0.354	−0.765	* 213	31	0.331	0.537	−0.646
* 721	38.000	0.378	0.622	−0.743	* 211	17	0.293	0.797	−0.604
513		0.357	0.030	−0.734	486	10	0.649	0.604	−0.594
* 337	9.000	0.187	0.260	−0.710	* 451	82	0.449	0.497	−0.569
313	9.000	0.242	0.277	−0.710	442	64	0.482	0.576	−0.567
515	17.000	0.338	0.176	−0.701	713	44	0.553	0.557	−0.566
711	27.000	0.211	0.553	−0.678	* 314	10	0.337	0.348	−0.554
* 213	31.000	0.171	0.534	−0.662	* 236	16	0.328	0.079	−0.546
* 451	82.000	0.074	0.271	−0.659	448	90	0.426	0.553	−0.531
238	16.000	0.169	0.190	−0.657	999		0.000	0.005	−0.524
* 211	17.000	0.228	0.701	−0.655	324	12	0.306	0.490	−0.520
* 236	16.000	0.273	0.026	−0.647	441	44	0.342	0.307	−0.517
331	9.000	0.166	0.305	−0.638	722	53	0.565	0.567	−0.516
321	9.000	0.209	0.287	−0.630	481	48	0.336	0.732	−0.514
* 323	14.000	0.204	0.244	−0.625	* 721	38	0.556	0.495	−0.512
519	10.000	0.058	0.158	−0.623	* 337	9	0.229	0.296	−0.492
492	14.000	0.154	0.347	−0.621	* 315	10	0.292	0.228	−0.490
421		0.190	0.086	−0.618	336	7	0.297	0.241	−0.480
322	10.000	0.263	0.426	−0.612	327	9	0.271	0.322	−0.460
* 315	10.000	0.120	0.115	−0.611	* 323	14	0.346	0.220	−0.460
Mean	19.333	0.212	0.298	−0.665	Mean	32	0.369	0.422	−0.533
Median	14.000	0.206	0.274	−0.656	Median	17	0.337	0.493	−0.522

Table 5

Stock returns, financial flexibility measures interacted with COVID-19 exposure measures, and stock and firm characteristics

Cross-sectional regressions of cumulative excess stock returns from February 3 to March 23, 2020, on financial flexibility measures interacted with an indicator for firms highly exposed to COVID-19 under each exposure measure. Controls include equity beta, the 2019 stock return, the book-to-market ratio, the log market value of equity, gross profitability, capex, SG&A, R&D, payouts, and an investment-grade rating indicator. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Affected share (TQ)	(2) Teamwork share (TQ)	(3) Customer share (TQ)	(4) Presence share (TQ)	(5) Q2 2020 year-on-year change in sales (BQ)	(6) Share of jobs that can be performed from home (BQ)	(7) Labor intensity based on number of employees (TQ)	(8) Manually classified as highly affected
Cash / assets × highly affected	0.203**	0.144*	0.232***	0.085	−0.039	0.059	−0.135*	0.179*
	(2.05)	(1.92)	(2.92)	(0.69)	(−0.53)	(0.55)	(−1.95)	(1.77)
St debt / assets × highly affected	0.303**	0.229*	−0.051	0.365**	0.069	0.060	0.141	0.392***
	(2.19)	(1.66)	(−0.39)	(2.46)	(0.54)	(0.43)	(0.97)	(2.78)
Lt debt / assets × highly affected	0.061	0.099*	0.133**	0.016	0.037	0.034	−0.095	0.041
	(1.06)	(1.68)	(2.35)	(0.26)	(0.68)	(0.58)	(−1.56)	(0.71)
Cash / assets	0.011	−0.003	−0.016	0.030	0.047	0.029	0.070*	0.014
	(0.29)	(−0.07)	(−0.42)	(0.83)	(1.26)	(0.78)	(1.71)	(0.38)
St debt / assets	−0.178**	−0.171**	−0.087	−0.185**	−0.081	−0.121	−0.150*	−0.198***
	(−2.36)	(−2.30)	(−1.12)	(−2.51)	(−0.96)	(−1.61)	(−1.90)	(−2.64)
Lt debt / assets	−0.159***	−0.174***	−0.188***	−0.147***	−0.152***	−0.157***	−0.125***	−0.154***
	(−4.86)	(−5.42)	(−5.78)	(−4.82)	(−4.93)	(−4.99)	(−4.09)	(−4.73)
Ind: Highly affected	−0.054	−0.102***	−0.067*	−0.002	−0.154***	−0.034	0.021	−0.079**
	(−1.51)	(−2.73)	(−1.83)	(−0.06)	(−5.43)	(−0.98)	(0.73)	(−2.19)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,211	2,211	2,211	2,211	2,177	2,228	2,189	2,211
Adjusted R^2	0.133	0.133	0.135	0.133	0.175	0.131	0.132	0.134

Table 6

Accounting performance (cumulative Q1+Q2), financial flexibility measures interacted with affectedness, and firm characteristics

Cross-sectional regressions of return on assets, defined as cumulative operating income before depreciation for the first two quarters of 2020 divided by total assets at year-end 2019, winsorized at the 1st and 99th percentiles. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) ROA	(2) ROA
Cash / assets × highly affected	0.150***	0.104***
	(3.31)	(3.26)
St debt / assets × highly affected	−0.007	−0.054
	(−0.11)	(−1.16)
Lt debt / assets × highly affected	−0.001	0.021
	(−0.05)	(1.15)
Cash / assets	−0.104***	−0.054***
	(−6.97)	(−4.39)
St debt / assets	−0.270***	−0.076***
	(−7.92)	(−2.80)
Lt debt / assets	0.001	−0.015
	(0.08)	(−1.48)
Ind: Highly affected	−0.041**	−0.040***
	(−2.40)	(−3.48)
Payout / assets		0.055*
		(1.90)
Ind: IG debt rating		−0.015**
		(−2.26)
Capex / lagged assets		0.021
		(0.57)
SGA / sales		−0.009***
		(−7.87)
COGS / sales		−0.016
		(−1.58)
RD / assets		−0.101***
		(−9.25)
Firm age		0.000*
		(1.66)
Lagged ROA: Q1+Q2 2019 / assets EoFY18	0.423***	0.401***
	(38.97)	(32.18)
Equity beta		−0.005
		(−1.38)
BM at EoY 2019	0.007	−0.005
	(1.64)	(−1.52)
ln(MVE) at EoY 2019	0.012***	0.007***
	(9.30)	(6.42)
Momentum 2019		0.004
		(1.32)
Observations	2,503	2,094
Adjusted R^2	0.645	0.683

Table 7

Payout ratios by quartile and financial flexibility measures

Financial flexibility measures and payout ratios for the entire sample and by payout-ratio quartile. (Cash + payout) / (assets + payout) is the cash-to-assets ratio had the firm retained its payouts; (Lt debt − payout) / assets is the long-term debt ratio had the firm used its payouts to repay long-term debt. The left panel uses 2019 payouts; the right panel uses cumulative 2017–2019 payouts over 2019 assets.

	2019 payouts / 2019 assets			(2017+2018+2019 payouts) / 2019 assets		
	N	Median	Mean	N	Median	Mean
All firms						
Cash / assets	2,729	0.130	0.257	2,649	0.129	0.254
(Cash + payout) / (assets + payout)	2,729	0.167	0.281	2,649	0.225	0.316
Lt debt / assets	2,722	0.213	0.253	2,642	0.217	0.255
(Lt debt - payout) / assets	2,722	0.176	0.163	2,642	0.123	0.082
Payout / assets	2,729	0.007	0.090	2,649	0.031	0.173
Top quartile payout ratio						
Cash / assets	683	0.107	0.176	663	0.117	0.193
(Cash + payout) / (assets + payout)	683	0.183	0.252	663	0.318	0.374
Lt debt / assets	680	0.254	0.288	660	0.249	0.282
(Lt debt - payout) / assets	680	0.169	−0.052	660	−0.030	−0.328
Payout / assets	683	0.076	0.339	663	0.239	0.609
Bottom quartile payout ratio						
Cash / assets	682	0.482	0.472	662	0.469	0.471
(Cash + payout) / (assets + payout)	682	0.482	0.472	662	0.469	0.471
Lt debt / assets	682	0.078	0.174	662	0.090	0.197
(Lt debt - payout) / assets	682	0.078	0.174	662	0.090	0.197
Payout / assets	682	0.000	0.000	662	0.000	0.000
Middle quartiles payout ratios						
Cash / assets	1,364	0.091	0.191	1,324	0.087	0.177
(Cash + payout) / (assets + payout)	1,364	0.104	0.200	1,324	0.128	0.209
Lt debt / assets	1,360	0.256	0.275	1,320	0.256	0.270
(Lt debt - payout) / assets	1,360	0.245	0.265	1,320	0.212	0.230
Payout / assets	1,364	0.007	0.011	1,324	0.030	0.040

Table 8

Stock returns, financial flexibility measures, and a top quartile payout ratio indicator variable

Cross-sectional regressions of excess stock returns on firm characteristics and an indicator equal to one if cumulative 2017–2019 payouts over 2019 assets are in the top quartile. Collapse period: February 3 to March 23, 2020. Stimulus day: March 24, 2020. All regressions include Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Collapse period	(2) Stimulus day	(3) Collapse period	(4) Stimulus day
Ind: 3-year cum. payout / assets top quartile	0.019	−0.002	0.024*	−0.004
	(1.47)	(−0.48)	(1.76)	(−0.76)
Cash / assets			0.048	0.014
			(1.33)	(1.07)
St debt / assets			−0.110*	−0.002
			(−1.67)	(−0.09)
Lt debt / assets			−0.151***	0.033***
			(−5.36)	(3.18)
Ind: IG debt rating			0.018	−0.013*
			(0.89)	(−1.77)
Capex / lagged assets			−0.101	0.068
			(−0.84)	(1.53)
SGA / sales			−0.003	0.001
			(−0.84)	(0.86)
COGS / sales			0.002	0.006
			(0.07)	(0.48)
RD / assets			0.028	−0.009
			(0.86)	(−0.78)
Equity beta	−0.049***	0.019***	−0.060***	0.014***
	(−5.12)	(5.36)	(−5.42)	(3.44)
BM at EoY 2019	0.012	−0.004	−0.001	−0.002
	(1.43)	(−1.38)	(−0.07)	(−0.51)
ln(MVE) at EoY 2019	0.004	0.007***	0.002	0.009***
	(1.43)	(7.20)	(0.43)	(6.45)
Momentum 2019	−0.017*	0.005*	−0.023**	0.008**
	(−1.91)	(1.77)	(−2.13)	(2.09)
Profitability	−0.004	0.003	−0.041	0.003
	(−0.28)	(0.48)	(−1.46)	(0.34)
Observations	2,623	2,616	2,205	2,200
Adjusted R^2	0.101	0.099	0.133	0.110

Table 9

Stock returns, financial flexibility measures, and conglomerate indicator variables

Cross-sectional regressions of excess stock returns on firm characteristics and an indicator for COVID-19 conglomerates: firms with segments in two or more NAICS three-digit industries, at least one but not all of which are highly affected. Regressions do not include industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Collapse period	(2) Stimulus day	(3) Collapse period	(4) Stimulus day
Cash / assets × Ind: COVID-19 conglomerate			0.141	−0.062
			(1.02)	(−1.23)
St debt / assets × Ind: COVID-19 conglomerate			0.865***	−0.074
			(3.57)	(−0.84)
Lt debt / assets × Ind: COVID-19 conglomerate			0.241***	−0.016
			(2.59)	(−0.48)
Cash / assets × Ind: Highly affected (0 for conglomerates)			0.194*	0.046
			(1.77)	(1.15)
St debt / assets × Ind: Highly affected (0 for conglomerates)			0.384***	−0.008
			(2.66)	(−0.16)
Lt debt / assets × Ind: Highly affected (0 for conglomerates)			0.064	0.057***
			(1.11)	(2.71)
Cash / assets	0.106***	0.012	0.077**	0.009
	(3.02)	(0.91)	(2.05)	(0.68)
St debt / assets	−0.059	−0.004	−0.203***	0.005
	(−0.90)	(−0.16)	(−2.61)	(0.16)
Lt debt / assets	−0.157***	0.043***	−0.189***	0.028**
	(−5.75)	(4.33)	(−5.66)	(2.32)
Ind: COVID-19 conglomerate	−0.008	−0.005	−0.139***	0.012
	(−0.38)	(−0.66)	(−3.05)	(0.73)
Ind: Highly affected (set to zero for COVID-19 conglomerate)	−0.058***	0.020***	−0.118***	−0.002
	(−3.96)	(3.84)	(−3.91)	(−0.21)
Ind: IG debt rating	0.007	−0.015**	0.009	−0.014*
	(0.34)	(−2.03)	(0.46)	(−1.88)
Capex / lagged assets	−0.335***	0.060	−0.313***	0.057
	(−3.00)	(1.47)	(−2.80)	(1.40)
SGA / sales	0.002	0.001	0.002	0.000
	(0.51)	(0.38)	(0.48)	(0.33)
COGS / sales	−0.063**	0.030***	−0.064**	0.030***
	(−2.15)	(2.81)	(−2.19)	(2.80)
RD / assets	0.021	−0.007	0.038	−0.008
	(0.63)	(−0.55)	(1.14)	(−0.62)
Payout / assets	−0.013	0.042	−0.021	0.037
	(−0.13)	(1.22)	(−0.22)	(1.06)
Equity beta	−0.068***	0.012***	−0.067***	0.013***
	(−6.50)	(3.27)	(−6.38)	(3.47)
BM at EoY 2019	−0.003	−0.000	−0.000	−0.000
	(−0.30)	(−0.06)	(−0.05)	(−0.09)
ln(MVE) at EoY 2019	0.006	0.008***	0.006*	0.008***
	(1.61)	(6.55)	(1.78)	(6.43)
Momentum 2019	−0.025**	0.012***	−0.026**	0.013***
	(−2.36)	(3.14)	(−2.44)	(3.29)
Profitability	0.016	0.002	0.019	0.003
	(0.61)	(0.25)	(0.73)	(0.30)
One-sided p: conglomerate > highly affected	0.010	0.999	0.665	0.217
Industry fixed effects	No	No	No	No
Observations	2,211	2,206	2,211	2,206
Adjusted R^2	0.083	0.085	0.090	0.087

Table 10

Stock returns and measures of financial constraints

Cross-sectional regressions of excess stock returns on indicators of financial constraints. Collapse period: February 3 to March 23, 2020. Stimulus day: March 24, 2020. All regressions include the asset pricing characteristics and Fama-French 49 industry fixed effects. t-statistics are in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

	(1) Collapse	(2) Stimulus	(3) Collapse	(4) Stimulus	(5) Collapse	(6) Stimulus	(7) Collapse	(8) Stimulus
FC (WW index)	0.024	0.001						
	(1.31)	(0.15)						
FC (KZ index)			−0.020	0.011**				
			(−1.55)	(2.43)				
FC (SA index)					0.028*	−0.004		
					(1.76)	(−0.73)		
FC (HR cash ex ante)							−0.011	−0.007
							(−0.85)	(−1.57)
Equity beta	−0.056***	0.020***	−0.052***	0.016***	−0.043***	0.017***	−0.049***	0.018***
	(−5.79)	(5.45)	(−5.37)	(4.57)	(−4.71)	(5.17)	(−5.23)	(5.37)
BM at EoY 2019	0.009	−0.003	0.014*	−0.005	0.012	−0.003	0.009	−0.003
	(1.15)	(−1.00)	(1.65)	(−1.61)	(1.52)	(−0.99)	(1.10)	(−1.18)
ln(MVE) at EoY 2019	0.006**	0.007***	0.004	0.007***	0.006**	0.007***	0.004	0.007***
	(2.08)	(6.82)	(1.34)	(7.17)	(2.04)	(6.71)	(1.41)	(7.38)
Momentum 2019	−0.017*	0.006*	−0.013	0.008**	−0.013	0.006*	−0.018**	0.005*
	(−1.81)	(1.78)	(−1.51)	(2.57)	(−1.58)	(1.91)	(−2.09)	(1.75)
Profitability	−0.000	0.004	0.001	−0.001	0.001	0.003	−0.005	0.002
	(−0.01)	(0.65)	(0.05)	(−0.10)	(0.09)	(0.64)	(−0.35)	(0.38)
Observations	2,474	2,467	2,579	2,572	2,692	2,684	2,667	2,660
Adjusted R^2	0.112	0.100	0.107	0.097	0.101	0.101	0.103	0.101

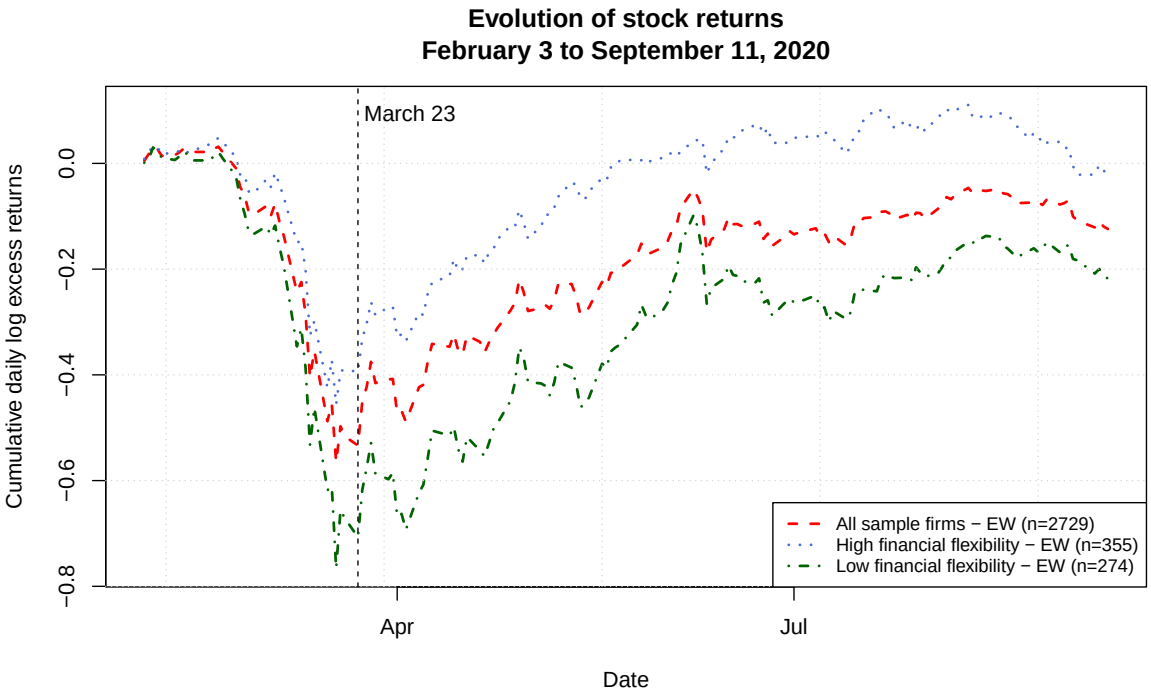


Figure 1

Evolution of stock returns for groups of firms with high and low financial flexibility

Cumulative daily log excess stock returns from February 3 to September 11, 2020, for an equal-weighted portfolio of all sample firms, a portfolio of firms with high financial flexibility (top quartile of cash over assets and bottom quartile of long-term debt over assets at the end of fiscal year 2019), and a portfolio of firms with low financial flexibility (bottom quartile of cash and top quartile of long-term debt).

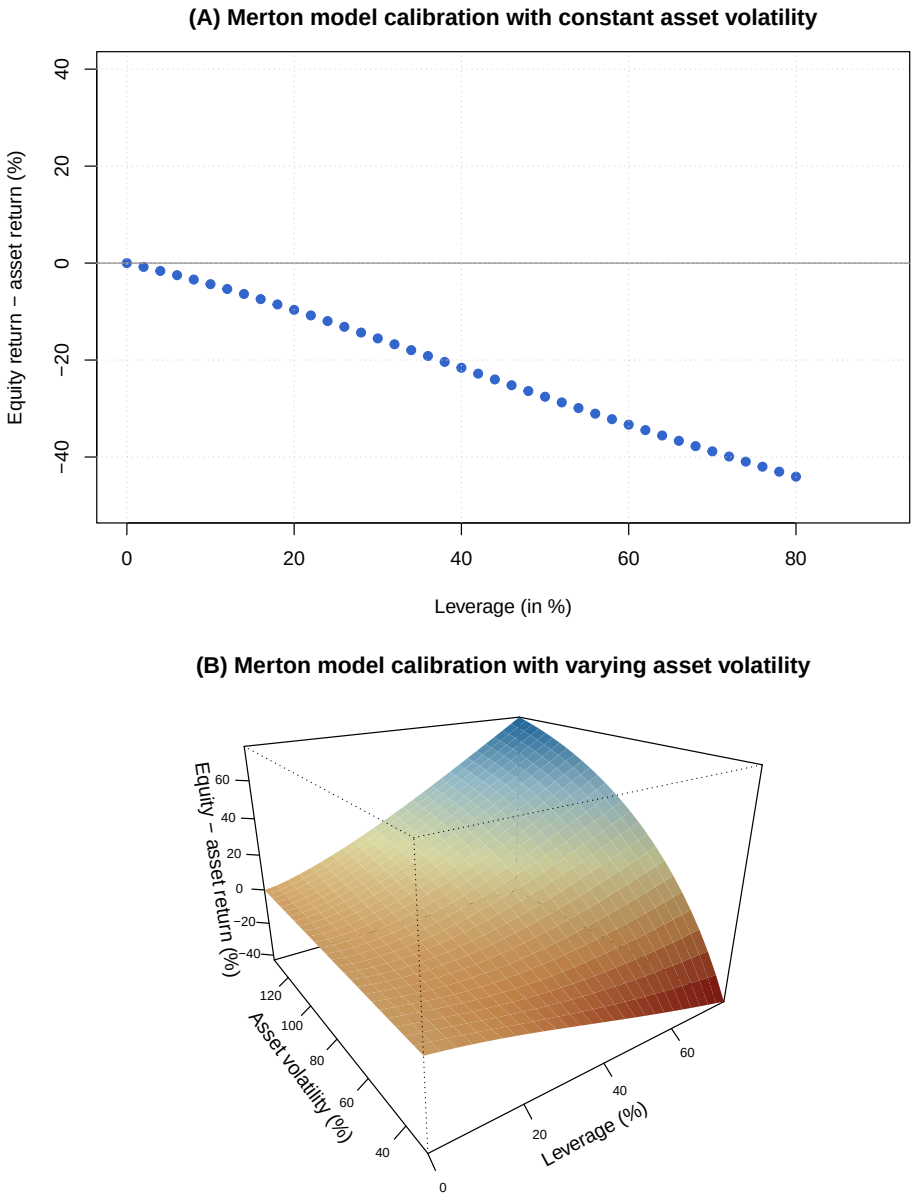


Figure 2
Calibration of the Merton model

Panel A shows the difference between equity and asset returns for a large shock to asset value as a function of leverage (the face value of debt) in a calibration of the Merton (1974) model, with $V_0 = 100$, $V_{Shock} = 70$, $\sigma_{assets} = 30\%$, $r = 3\%$, $q = 1\%$, and $T = 5$ years. Panel B shows the same difference as a function of both leverage and asset volatility.